Qubit Chat: A Controlled Evaluation of Grover-Inspired Reranking for Document Retrieval

Anishitha Kovvuri, Bala Rakesh Gundapaneni, Abdul Basith Syed,
Jagadeeshbabu Borru, Naga Sravanthi Puppala
Department of Computer Science and Engineering
SRM University AP, Andhra Pradesh 522502, India
anishitha_kovvuri@srmap.edu.in, balarakesh_gundapaneni@srmap.edu.in
abdulbasith_syed@srmap.edu.in, jagadeeshbabu_borru@srmap.edu.in
Corresponding author: nagasravanthi.p@srmap.edu.in

***Abstract*—Selecting useful retrieval evidence requires balancing relevance with repeated information. We evaluate L6-guided Grover Adaptive Search as document-set selection after classical dense and lexical retrieval. A cross-encoder supplies relevance scores, and a compact ideal simulator performs phase marking and amplitude amplification over improving subsets. The base policy selects eight of fifteen candidates. A constrained refinement expands the candidate pool, preserves the first five results, and searches for a less redundant three-document tail under neural-score constraints. Full tests cover scientific claims, nutrition-related retrieval, and financial questions. FiQA currently uses the earlier base configuration. The reported full-test normalized discounted cumulative gain at rank five is 0.6878 on SciFact, 0.3689 on NFCorpus, 0.3575 on FiQA. SciFact redundancy falls 3.21 percent relative to matched base selection. NFCorpus redundancy falls 2.63 percent relative to matched base selection. Matched classical refiners attain the same aggregate quality on those follow-up tests, and the extra neural scoring increases complete pipeline latency. We provide paired query statistics, classical and neural controls, independent metric checks, oracle accounting, and mathematical preservation guarantees. These results characterize a relevance–redundancy tradeoff without establishing physical quantum acceleration, lower monetary cost, or improved generated answers.**

***Index Terms*—Grover Adaptive Search, document retrieval, cross-encoder, evidence selection, redundancy.**

I. Introduction

Retrieval-augmented systems retrieve documents before producing an answer [1]. A limited evidence budget makes both relevance and overlap important: several near-duplicate passages can consume space that might otherwise cover a different aspect of a question. Independent document scores do not explicitly consider relationships among selected documents. Set selection instead evaluates an evidence bundle as a whole, but its benefit must be measured against strong relevance models and classical methods optimizing the same objective.

Grover search reduces oracle queries under a specified access model [2], [3]. It does not remove classical embedding, retrieval, state preparation, or reversible computation costs. In a threshold-marking reranker, equally marked documents have symmetric ideal amplification. Their measurement probabilities therefore do not themselves provide a graded relevance ordering. Transforming those probabilities into ranking scores requires controls for both threshold fallback and relevance distortion.

We retain that original policy as a legacy comparator and use Grover Adaptive Search (GAS) to search document subsets instead [4]. A neural cross-encoder supplies relevance evidence, while document embeddings measure redundancy. An earlier base policy jointly optimizes relevance and redundancy over fifteen candidates. The follow-up refinement protects its five highest-ranked outputs, expands the available tail candidates, and minimizes redundancy subject to neural-score constraints. Grover amplification affects the sampled subset; it is not substituted for a learned relevance score.

Our questions are: (1) can constrained selection reduce redundancy while preserving the initial ranking prefix and limiting observed recall loss; (2) how does it compare with matched greedy, exact, random, lexical, and neural methods; and (3) what simulation work and resource assumptions are required? The contribution is an evaluated retrieval formulation and reproducible accounting, rather than a new quantum search algorithm. Gains shared with classical optimizers support the selection policy, not a uniquely quantum benefit. Redundancy is an embedding-similarity proxy, not a duplicate percentage, judged coverage score, or answer-quality measure.

II. Related Work

Amplitude amplification separates oracle queries from the cost of constructing and executing each oracle [2], [3]. Search with an unknown marked-state count motivates randomized iteration schedules [5]; quantum minimum finding updates incumbent thresholds [6]. Gilliam, Woerner, and Gonciulea formulate GAS for constrained polynomial binary optimization and describe arithmetic oracles [4]. Our compact simulator reproduces ideal phase and reflection operations over a small subset basis. It does not realize all corresponding decision, arithmetic, and constraint registers on a quantum device.

Fixed-cardinality selection can be represented with fixed-weight bit strings. Dicke-state constructions show that such states need not always require arbitrary dense state synthesis [7]. However, a preparation method's theoretical circuit bound cannot be assigned to an implementation that does not exe-

TABLE I
DATASET COUNTS AND FROZEN BASE LEXICAL/DIVERSITY WEIGHTS.

Dataset	Parents	Dev.	Test	β	λ
SciFact	5,183	161	300	0.25	0.000
NFCorpus	3,633	324	323	0.50	0.050
FiQA	57,638	500	648	0.00	0.025

cute it. Data access, arithmetic, and constraint checks remain central to interpreting quantum claims [8].

Maximal marginal relevance (MMR) balances relevance with novelty relative to previously selected documents [9]. Reciprocal-rank fusion (RRF) combines retrieval lists [10], and BM25 supplies lexical relevance [11]. Exact and greedy controls distinguish an objective's value from a search mechanism's value. Our matched refinement controls share the expanded candidates, fixed prefix, score constraints, and redundancy definition.

Sentence encoders provide reusable dense vectors [12]. Cross-encoders jointly encode a query and passage to estimate relevance [13]. The common `all-MiniLM-L6-v2` embedding model and `ms-marco-MiniLM-L6-v2` cross-encoder have different roles and training objectives [14], [15]. "MS MARCO" identifies the latter's passage-ranking training task, rather than an alternative generic encoder. BEIR supplies heterogeneous collections for testing retrieval across domains [16].

III. DATA AND EXPERIMENTAL PROTOCOL

A. Datasets and representations

Official BEIR SciFact, NFCorpus, and FiQA-2018 cover scientific claims, nutrition-related retrieval, and financial questions [16]. Table I reports parent-document and judged-query counts. Every positively judged official test query is included. Input checksums, query/document references, and development/test ID separation are verified. The larger FiQA collection extends corpus and domain coverage; it does not enlarge the quantum pool to the corpus size.

Dense representations use the pinned encoder `sentence-transformers/all-MiniLM-L6-v2`, revision `1110a243...529b4d41`, with 384-dimensional normalized vectors. Text uses Unicode NFKC and whitespace normalization. Full tokenizer offsets are obtained with padding and truncation disabled, including for the first document. Documents use at most 190 WordPiece content tokens per chunk with a step of 158; two special tokens yield at most 192 model-input tokens. Parent relevance is maximum query cosine over chunks. The representative passage is the maximizing original chunk, with the lowest chunk index breaking exact ties. Methods share parent IDs and representative passages for a given pool.

Queries are checked against the 190-content-token limit. Cross-encoder pairs are checked against 512 tokens without silent truncation. The relevance model is `cross-encoder/ms-marco-MiniLM-L6-v2`, revision `233902d2...bee0a`; the L2 control uses revision `1b5cd67b...9b787`. Full revisions and model-file hashes are retained in provenance. No test judgments enter retrieval, neural inference, or selection.

B. Development choices and follow-up protocol

SciFact development is a deterministic 20% subset of official training queries; NFCorpus and FiQA use official development splits. The base lexical grid is $\beta \in \{0, 0.25, 0.5\}$, and its diversity grid is $\lambda \in \{0, 0.01, 0.025, 0.05\}$. All stochastic settings use seeds 0–4. Eligibility requires development mean nDCG@5 and Recall@8 to equal or exceed the same-shortlist L6-only baseline within 10^{-12}. Eligible settings prefer higher nDCG, higher recall, lower redundancy, smaller penalty, and smaller lexical weight, in that order. Zero penalty is retained when it wins.

L6 MMR is independently tuned over $\{0, 0.1, 0.25, 0.5, 0.75, 1\}$, prioritizing nDCG, recall, lower redundancy, and smaller weight. Its selected weight is zero on these collections. This is a relevance-prioritizing development choice, not the entire MMR tradeoff frontier. Cosine-guided controls retain separate frozen development settings. Legacy thresholds $\{0.3, 0.5, 0.7\}$ are selected on development nDCG@5 using matched classical sampling; ties prefer 0.5, then 0.3, then 0.7.

Refinement uses deterministic 100-query training confirmation with namespace `accuracy-v3-confirmation-v1`, excluding historical development, test, and pilot IDs. Its gate requires unchanged mean nDCG@5, at least 98% of matched base mean Recall@8, and strictly lower mean redundancy. The recall limit is relative, not two absolute percentage points. These mean gates do not prove statistical noninferiority. NFCorpus's 100-query training confirmation failed the recall gate (-2.1561% relative change). Full-test collection nevertheless proceeded as an explicitly recorded protocol exception with unchanged ranking parameters; its subsequent test result does not erase that failure.

SciFact and NFCorpus include protected-prefix refinement; FiQA reports the earlier fifteen-candidate configuration. Matched optimizers use the corresponding configuration; these are distinct policies within the L6-guided GAS family. The newer official tests are follow-up evaluations: earlier test outcomes had already informed method development. They are not an untouched holdout for the redesigned policy. The recorded parameters are not retuned against follow-up judgments. Training-confirmation outcomes are not substituted for full-test effectiveness.

C. Comparators, metrics, and timing

We retain twenty original methods and add seven expanded-pool comparisons where refinement is evaluated. Controls include cosine, BM25/dense RRF with rank constant 60, native cosine and L6 MMR, cosine-guided GAS, L6-only ranking, base L6-guided GAS, greedy/exact optimizers, random search, legacy Grover, and L2/L6 cross-encoders. Dense-fifteen and dense-sixty-four neural controls return eight in cross-encoder

order. Their pools differ from hybrid-fifteen L6-only and the expanded union. Comparisons characterize complete policies, not isolated search substitutions.

The seven expanded-pool methods are constrained GAS, local-swap greedy, exact float, exact quantized, matched random search, global union-L6, and protected-prefix union-L6. Matched refiners inherit each seed's baseline prefix. Random refinement draws $\max(1, 2q)$ uniform subsets, where q is that seed's actual refinement phase count, plus its initial incumbent. This matches a declared objective-access allowance, not exhaustive simulator work. Exact controls independently enumerate their objectives; GAS receives neither their optimum nor marked-state counts.

Metrics are nDCG@5, Recall@8, and redundancy; nDCG@8 diagnoses changes below the protected prefix. nDCG uses linear gains and logarithmic discount, checked against NIST `trec_eval` [17]. Five seed values are averaged within each query, then across queries; seeds are not independent queries. Complete method exports accompany the compact tables.

The host is an Intel Core i7-1355U, 12 logical CPUs, approximately 15.3 GiB RAM, Linux, and Python 3.12.3. Processes use two compute threads. Packages include NumPy 1.26.4, SciPy 1.13.1, PyTorch 2.12.0+cpu, Transformers 4.57.6, Sentence Transformers 3.4.1, Qiskit 0.46.0, and Aer 0.13.3. Data/model restore and artificial-query warm-up are separate. Shared-host load and session boundaries are retained; the machine is not a dedicated latency laboratory.

Online latency composes query encoding, dense retrieval, representative features, applicable lexical processing, original/additional neural inference, baseline selection, and refinement. Shared upstream measurements are charged to each applicable policy. Selector latency excludes retrieval/inference but includes objective preparation and bookkeeping. Combined selector percentiles add actual baseline and refinement times per request before aggregation. P95 is over seeded requests, never divided by five. Across-session differences are not matched speed experiments.

IV. L6-Guided Grover Set Selection

A. Retrieval and base objective

Let $c_i \in [-1, 1]$ be parent query cosine among dense top 64 and $b_i \geq 0$ its BM25 score. The reproduced hybrid score is

$$h_i = (1-\beta)\frac{1+c_i}{1+\max_j c_j} + \beta\frac{b_i}{\max_j b_j}. \tag{1}$$

If lexical scores are negligible, use only the dense term. If its denominator is negligible, use $(1 + c_i)/2$. Negligibility uses tolerance 10^{-12}. Hybrid top fifteen is sorted by score and document ID. Keep raw logits z_i, cosine, embeddings, and normalized relevance separate. Within the original fifteen,

$$r_i = \frac{z_i - \min_j z_j}{\max_j z_j - \min_j z_j}, \tag{2}$$

with $r_i = 0.5$ for negligible range. Negative logits are valid.

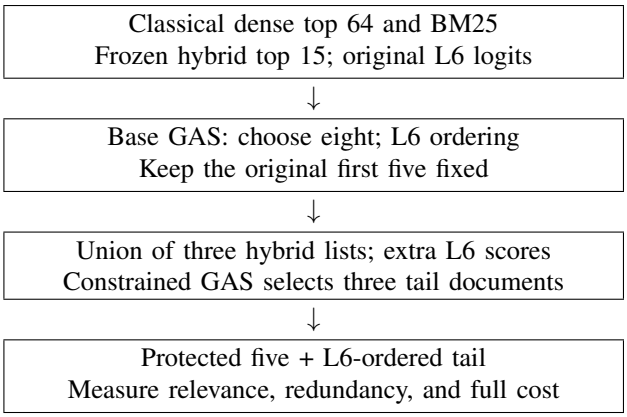


Fig. 1. Base selection and protected-prefix refinement. The base configuration returns after the second stage; refinement cost includes every applicable upstream stage.

For normalized representative vectors v_i, define $d_{ij} = \mathrm{clip}(v_i^\top v_j, 0, 1)$. For $|S| = k = 8$,

$$\mathcal{R}(S) = \binom{k}{2}^{-1} \sum_{\{i,j\}\subset S} d_{ij}, \tag{3}$$

$$U(S) = k^{-1} \sum_{i\in S} r_i - \lambda \mathcal{R}(S). \tag{4}$$

The base uses eight-bit coefficients, $a = 255$, and energy

$$q_i = \mathrm{round}(a r_i), \quad q_{ij} = \mathrm{round}\left(\frac{2a\lambda}{k-1} d_{ij}\right), \tag{5}$$

$$E_b(S) = -\sum_{i\in S} q_i + \sum_{\{i,j\}\subset S} q_{ij}. \tag{6}$$

Rounding is ties-to-even. Raw-L6 top eight initializes the incumbent; raw L6 descending with document-ID ties determines final order. Summing coefficient errors gives, for each fixed set,

$$\left| \frac{-E_b(S)}{ak} - U(S) \right| \leq \frac{k+1}{4a}. \tag{7}$$

This bounds approximation, not retrieval accuracy. At zero penalty the top-L6 incumbent maximizes relevance and is retained; quantization does not justify arbitrary equal-energy reordering.

B. Protected-prefix refinement

Figure 1 shows how the base feeds refinement. Form union C of hybrid top-fifteen lists at $\beta = 0, 0.25, 0.5$, within the same dense top 64. It has at most 45 documents. Score the original shortlist first and preserve its exact logits; score only additional candidates afterward. This prevents neural batching changes from altering the base result.

For each seed, let base ranking be $(s_1, \ldots, s_8)$, fixed prefix $H = (s_1, \ldots, s_5)$, and original tail T_0. Search $T \subseteq C \setminus H$, $|T| = 3$. Ascending sorted tail logits must satisfy

$$z_{(j)}(T) \geq z_{(j)}(T_0) - \varepsilon, \quad j = 1, 2, 3, \quad \varepsilon = 10^{-12}. \tag{8}$$

Candidates below the smallest original-tail logit minus ε can safely be discarded: they cannot occupy even the lowest

position. Let M be the remaining tail-pool size. Constraint (8) is equivalent to requiring at least $4 - j$ tail scores above each fixed bound $z_{(j)}(T_0) - \varepsilon$, using inclusive comparisons. A count violation identifies a sorted position below its bound, and meeting every count constraint places each position at or above its bound. This also handles tied scores.

Minimize $\mathcal{R}(H \cup T)$, including prefix–prefix, prefix–tail, and tail–tail pairs. The fixed prefix term is constant but remains in reported redundancy. With $a = 2^{16} - 1$, refinement energy is

$$E_r(H \cup T) = \sum_{\{i,j\} \subset H \cup T} \mathrm{round}(a d_{ij}). \tag{9}$$

Mark valid states with strictly lower integer energy than the incumbent. Accept a measured update only if true redundancy also decreases by more than ε. Output the original prefix followed by the new tail in raw-L6 order. This deliberately differs from a global L6 sort when an added document scores above a protected one.

C. Guarantees and counterexample

Prefix preservation. With fixed judgments, nDCG@5 depends only on the first five IDs and their order. Preserving them keeps DCG@5 and its ideal denominator identical. Per-query nDCG@5 and its seed/query means are therefore unchanged by construction. This is not an empirical accuracy gain uniquely due to Grover search; nDCG@8 and Recall@8 can change.

Feasibility and redundancy. The unchanged T_0 satisfies (8), survives the prefilter, and appears in the size-three basis. It supplies eight unique outputs even at budget exhaustion or when no improvement is sampled. Every accepted update strictly lowers true redundancy. Induction therefore gives final redundancy no greater than base redundancy. Strict improvement is not guaranteed per query; a trivial basis or an optimum can retain the original set.

Quantization bound. Each of the 28 rounded pair coefficients differs from its unrounded value by at most $1/2$. Thus

$$\left| \frac{E_r(S)}{28a} - \mathcal{R}(S) \right| \le \frac{1}{2a}. \tag{10}$$

Applying this twice bounds exact quantized-optimizer redundancy regret by $1/a$ relative to the exact float optimizer over the same admissible domain. It does not prove budget-limited GAS reaches that optimizer. At sixteen bits the per-set bound is approximately 7.63×10^{-6}; model error is not bounded by coefficient precision.

Recall counterexample. Replacing a relevant original-tail document by an irrelevant but higher-L6-scoring document can satisfy (8) while reducing recall. Therefore score dominance does not guarantee true relevance. Empirical recall gates and paired uncertainty remain necessary; neither an unchanged prefix nor higher logits establish overall noninferiority.

D. Amplitude amplification and budgets

The base has $\binom{15}{8} = 6{,}435$ cardinality-feasible subsets. Refinement has $F = \binom{M}{3}$ size-three subsets after score prefiltering, including subsets later failing (8). Only valid improving states are marked. For $|u\rangle = F^{-1/2} \sum_x |x\rangle$, one step is

$$G = (2|u\rangle\langle u| - I) O_\tau, \tag{11}$$

$$O_\tau |x\rangle = (-1)^{[\mathrm{valid}(x) \wedge E(x) < \tau]} |x\rangle. \tag{12}$$

The simulator explicitly flips marked amplitudes and reflects amplitudes about their post-oracle mean. These operations preserve the norm; measurement samples squared amplitudes. This is ideal amplitude amplification executed classically, not execution on quantum hardware.

Each stage begins with its incumbent and window $w = 1$. A trial draws an integer from zero through $\lceil w \rceil - 1$, limited by remaining phase calls, prepares the uniform state, applies that many steps, and measures. Strict improvement resets w; otherwise it grows by 1.2, capped at $\sqrt{F}$. Each stage allows 64 phase calls, 32 trials, or a cooperative 15-second limit per seed. Baseline and refinement budgets are separate: a refined request can use up to 128 phase calls. The controller receives no optimum or marked-state count. Trivial bases and every termination reason remain recorded.

TABLE II
REPORTED L6-GUIDED GAS FULL-TEST MEANS. RED. IS LOWER-IS-BETTER. SCIFACT AND NFCORPUS INCLUDE PROTECTED-PREFIX REFINEMENT; FIQA REPORTS THE EARLIER FIFTEEN-CANDIDATE CONFIGURATION. MATCHED OPTIMIZERS USE THE CORRESPONDING CONFIGURATION; THESE ARE DISTINCT POLICIES WITHIN THE L6-GUIDED GAS FAMILY.

Dataset	nDCG@5	nDCG@8	Recall@8	Red.↓
SciFact	0.687833	0.698173	0.801111	0.433636
NFCorpus	0.368865	0.346506	0.152997	0.431771
FiQA	0.357490	0.376939	0.434063	0.531244

V. RESULTS AND INTERPRETATION

A. Effectiveness and controls

Table II gives reported L6-guided GAS results, and Table III provides retrieval/neural comparisons. SciFact preserved nDCG@5, changed mean Recall@8 by -0.93% relative, and reduced redundancy by 3.21% relative to the base GAS measured in the same run. Matched greedy and exact-float selection attained the same aggregate effectiveness and redundancy. NFCorpus preserved nDCG@5, changed mean Recall@8 by +1.95% relative, and reduced redundancy by 2.63% relative to the base GAS measured in the same run. Matched greedy and exact-float selection attained the same aggregate effectiveness and redundancy. FiQA's full-test base GAS reached 0.3575 nDCG@5 and 0.4341 Recall@8; its earlier configuration is not evidence for protected-prefix refinement there. In the protected setting, unchanged nDCG@5 follows from the prefix proof. Tail membership changes explain why reporting only that protected metric would conceal recall and nDCG@8 tradeoffs.

The matched classical controls distinguish a selection objective's benefit from a search mechanism's benefit. Exact

TABLE III

FULL-TEST COMPARISONS. N@5 IS NDCG@5, R@8 IS RECALL@8, AND RED. IS CLIPPED-POSITIVE COSINE REDUNDANCY. REPORTED GAS AND MATCHED OPTIMIZERS FOLLOW TABLE II'S CONFIGURATION NOTE. BASE GAS/L6-ONLY USE HYBRID FIFTEEN; DENSE-FIFTEEN NEURAL CONTROLS HAVE A DIFFERENT SHORTLIST.

	SciFact			NFCorpus			FiQA		
Method	n@5	R@8	Red.	n@5	R@8	Red.	n@5	R@8	Red.
L6-guided GAS	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5312
Base L6-guided GAS	0.6878	0.8086	0.4480	0.3689	0.1501	0.4434	0.3575	0.4341	0.5312
Cosine	0.6361	0.8111	0.4756	0.3181	0.1335	0.4398	0.3470	0.4153	0.5363
RRF	0.6677	0.8149	0.4727	0.3491	0.1394	0.4394	0.3509	0.4262	0.5327
Cosine-guided GAS	0.6311	0.7742	0.4490	0.3160	0.1235	0.3996	0.3452	0.4093	0.5215
Legacy Grover (64)	0.6364	0.8111	0.4756	0.3181	0.1334	0.4398	0.3458	0.4150	0.5363
L6 only (hybrid 15)	0.6878	0.8086	0.4480	0.3689	0.1500	0.4447	0.3575	0.4341	0.5313
L6 MMR (hybrid 15)	0.6878	0.8086	0.4480	0.3689	0.1500	0.4447	0.3575	0.4341	0.5313
Matched greedy	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5311
Matched exact float	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4343	0.5309
Matched exact integer	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5311
Matched random search	0.6878	0.8011	0.4338	0.3689	0.1531	0.4320	0.3575	0.4341	0.5313
L2 (dense 15)	0.6431	0.8049	0.4642	0.3509	0.1396	0.4486	0.3319	0.4217	0.5295
L6 (dense 15)	0.6790	0.8104	0.4655	0.3621	0.1423	0.4537	0.3575	0.4341	0.5313
L2 (dense 64)	0.6321	0.7916	0.4329	0.3481	0.1438	0.4468	0.2973	0.3918	0.4962
L6 (dense 64)	0.6815	0.8087	0.4379	0.3670	0.1437	0.4586	0.3535	0.4160	0.5033

enumeration is practical because only three tail positions change and score constraints often leave little freedom. Where greedy or exact selection matches GAS, improvement over base selection does not identify a Grover-specific gain. Global union-L6 and protected-prefix union-L6 controls, retained in the complete export, separate candidate expansion from prefix preservation. Some simpler controls attain lower redundancy with lower relevance; lower similarity alone is not universal policy superiority.

B. Paired statistics and recall

Table IV reports paired query differences. Confidence intervals use 10,000 paired bootstrap resamples with seed zero; sign-randomization tests use 10,000 permutations with seed one. Differences within 10^{-12} are numerical ties. Protected-run Holm correction covers nine comparators by four metrics per dataset, not just displayed rows. The base-only nDCG audit uses all six declared comparators.

Read recall changes with their intervals and query-level losses, not the mean gate alone. A positive mean change can coexist with harmed queries. An interval containing zero is not proof of equivalence or noninferiority [18]. Numerical averaging artifacts do not establish significance. Companion evidence retains all comparison families, query increase/decrease counts, and nDCG@8 diagnostics.

C. Latency and search outcomes

Table V charges all online stages. Expanded neural scoring increases complete latency against matched base GAS. Refinement-only milliseconds cannot be presented as the cost of the complete two-stage method. Reduced neural inference may cost less than dense-sixty-four reranking, but candidate opportunities differ. No universal speed advantage follows. Physical quantum runtime, monetary expenditure, and energy are not measured.

Table VI distinguishes optimization from retrieval quality: an optimum for constrained redundancy can lose recall. Trial/budget termination retains a valid incumbent. Same-set counts include trivial bases and unsuccessful improvement attempts, distinct from legacy no-marked fallback. Complete evidence retains phase calls, measured trials, accepted updates, optimum gaps, and termination counts, without excluding unsuccessful queries or seeds.

D. Threshold and resource evidence

Table VII summarizes recorded development threshold comparisons; no-marked rates use saved cosine candidates. No new rankings or test-driven threshold selection are performed. Lower thresholds reduce the no-marked condition but can change ranking behavior; the frozen development winner is 0.7. Subset search instead marks adaptive incumbent-energy improvement. It too may have no improving states, but safely returns its incumbent rather than assigning relevance from an empty marked set.

Compact simulation enumerates all F basis energies before amplification. These simulator evaluations are separate from initial/measured incumbent checks and logical phase calls. A phase call is recorded as two reversible objective calls for compute/uncompute; constraints and preparation would add physical work. Because the simulator already has every energy, a classical argmin can solve the encoded problem without amplitude simulation. Its phase budget cannot establish wall-clock acceleration over exact enumeration.

Table VIII records observed pool/basis sizes. Table IX illustrates candidate scaling at fixed tail size three. Complex128 amplitudes occupy $16\binom{M}{3}$ bytes, excluding energy tables, layouts, embeddings, and model/process memory. A dense physical-register statevector instead needs $16\,2^q$ bytes for q qubits. These representation sizes are not interchangeable.

TABLE IV

REPORTED GAS MINUS MATCHED BASE GAS FOR PROTECTED RUNS; *BASE GAS MINUS L6-ONLY FOR THE EARLIER CONFIGURATION. INTERVALS ARE POINTWISE 95% INTERVALS; REJECTION DECISIONS USE THE FULL RECORDED HOLM FAMILY.

Dataset	Metric	Δ	95% paired CI	Raw p	Holm reject
SciFact	nDCG@5	+0.0000	[+0.0000, +0.0000]	1.0000	No
SciFact	nDCG@8	-0.0027	[-0.0081, +0.0023]	0.3043	No
SciFact	Recall@8	-0.0075	[-0.0242, +0.0075]	0.4565	No
SciFact	Redundancy	-0.0144	[-0.0168, -0.0122]	0.0001	Yes
NFCorpus	nDCG@5	+0.0000	[+0.0000, +0.0000]	1.0000	No
NFCorpus	nDCG@8	+0.0009	[-0.0025, +0.0045]	0.5897	No
NFCorpus	Recall@8	+0.0029	[+0.0004, +0.0062]	0.0369	No
NFCorpus	Redundancy	-0.0116	[-0.0140, -0.0094]	0.0001	Yes
FiQA	nDCG@5*	+0.0000	[+0.0000, +0.0000]	1.0000	No

TABLE V

COMPOSED ONLINE LATENCY IN MILLISECONDS PER SEEDED REQUEST. P50/P95 INCLUDE APPLICABLE RETRIEVAL, LEXICAL, NEURAL, AND SELECTION STAGES; LOADING/WARM-UP ARE SEPARATE. COMPARE WITHIN A DATASET/RUN, NOT SESSIONS AS MATCHED SPEED EXPERIMENTS.

	SciFact		NFCorpus		FiQA	
Method	P50	P95	P50	P95	P50	P95
L6-guided GAS	1406.5	1672.3	1146.8	1481.3	653.8	826.7
Base L6-guided GAS	963.2	1093.7	859.8	909.1	653.8	826.7
Cosine	33.3	48.3	24.5	32.9	79.9	98.5
RRF	61.6	94.9	31.0	44.5	225.3	328.3
Cosine-guided GAS	42.3	56.8	33.7	42.0	84.1	102.5
Legacy Grover (64)	38.2	313.6	25.2	281.6	84.3	231.9
L6 only (hybrid 15)	955.3	1085.4	851.9	901.4	650.0	821.1
L6 MMR (hybrid 15)	955.5	1085.7	852.1	901.6	650.1	821.3
Matched greedy	1405.4	1670.7	1145.5	1479.6	650.0	821.2
Matched exact float	1405.2	1670.6	1145.3	1479.3	651.2	822.5
Matched exact integer	1405.2	1670.6	1145.3	1479.3	651.1	822.4
Matched random search	1405.2	1670.6	1145.4	1479.3	651.9	824.8
L2 (dense 15)	348.0	403.0	310.5	331.5	238.9	294.6
L6 (dense 15)	915.8	1029.2	832.5	882.5	539.6	618.2
L2 (dense 64)	1374.7	1562.5	1242.5	1319.6	763.7	912.2
L6 (dense 64)	3799.0	4322.2	3460.8	3630.9	2037.9	2437.9

TABLE VI

REQUESTS AND MEAN BASE/REFINEMENT PHASE CALLS (BASE-ONLY SHOWS 0/BASE). HIT: EXACT FLOAT-REDUNDANCY OPTIMUM FOR REFINEMENT, EXACT INTEGER OPTIMUM FOR BASE-ONLY. SAME SET: RETAINED BASE MEMBERSHIP. S50/S95: COMBINED SELECTOR MS, EXCLUDING INFERENCE. LEGACY FALLBACK COUNTS HAVE THEIR OWN REQUEST DENOMINATOR.

Dataset	Requests	Base/refine calls	Hit	Same set	Legacy fallback	S50	S95
SciFact	1500	64.0/20.1	1.000	710	995/1500	9.66	10.67
NFCorpus	1615	63.9/17.8	1.000	957	1355/1615	9.49	11.02
FiQA	3240	0.0/64.0	0.994	–	2135/3240	3.95	4.54

Since $0 \le E_r \le 28(2^b - 1)$, a conservative signed register for incumbent difference has width

$$w_b = 1 + \mathrm{bitlength}\big(28(2^b - 1)\big)\,. \tag{13}$$

A fixed-weight decision representation adds M decision qubits; reversible score-count constraints require more workspace. Precision increases arithmetic widths and tightens (10), but does not reduce the amplitude count or NumPy integer storage width. Gate/depth costs depend on arithmetic, connectivity, constraints, preparation, and synthesis. A complete native circuit-depth sweep for the new constrained oracle was not performed. Small phase/reflection and reference-circuit checks establish limited numerical correctness, not deployment scaling. Legacy-oracle depths are not assigned to this different oracle.

VI. LIMITATIONS AND REPRODUCIBILITY

The classical top-64 and representative-chunk policies can exclude useful evidence before selection. Expansion stays within that pool; neural scoring supplies the relevance signal. A fixed quantum pool does not demonstrate quantum corpus search. The L2/L6 controls are concrete neural baselines, not every modern reranker. Development-selected zero-diversity

TABLE VII
LEGACY DEVELOPMENT THRESHOLDS. N@5 USES STORED MATCHED-SAMPLING OUTCOMES; NONE IS THE PERCENTAGE OF SAVED QUERIES WITH NO COSINE CANDIDATE MEETING THE THRESHOLD. QUERY-LEVEL RATES DO NOT TREAT SEEDS AS INDEPENDENT OBSERVATIONS.

	SciFact dev.		NFCorpus dev.		FiQA dev.	
τ_c	n@5	None (%)	n@5	None (%)	n@5	None (%)
0.3	0.6627	0.0	0.3010	0.6	0.3514	0.0
0.5	0.6579	6.8	0.3011	31.8	0.3515	1.8
0.7	0.6656	67.1	0.3026	87.0	0.3516	69.4

TABLE VIII
OBSERVED PROTECTED-REFINEMENT RESOURCE RANGES OVER SEEDED REQUESTS. REFINE MS IS MEDIAN REFINEMENT-ONLY TIME, NOT ONLINE OR COMBINED SELECTION TIME.

Dataset	Union	Tail pool	Basis states	Refine ms
SciFact	16–28	3–10	1–120	1.95
NFCorpus	15–30	3–16	1–560	1.87

TABLE IX
ANALYTICAL SCALING, NOT CIRCUIT EXECUTION MEASUREMENTS. w_b IS A CONSERVATIVE SIGNED OBJECTIVE-DIFFERENCE REGISTER AT b PAIR-COEFFICIENT BITS, EXCLUDING CONSTRAINT/PREPARATION ANCILLAS.

Tail pool	$\binom{M}{3}$	Amp. bytes	w_8	w_{12}	w_{16}
5	10	160	14	18	22
10	120	1,920	14	18	22
15	455	7,280	14	18	22
20	1,140	18,240	14	18	22
40	9,880	158,080	14	18	22

MMR does not establish superiority over its full operating curve.

Protected nDCG@5 is relative to a particular baseline prefix, not a globally optimal ranking. The recall gate is empirical, individual queries can worsen, and the configuration coverage differs as stated in Table II. Test reuse and the confirmation exception limit confirmatory inference. Embedding similarity does not directly measure semantic duplication, complementary evidence, hallucination, or answer quality.

The simulator is noiseless and exhaustively prepares objectives. Efficient reversible data loading, a full constrained oracle, a noise study, monetary/energy models, and generated-answer experiments are not demonstrated. Logical estimates exclude some workspace. Shared-host variation and different candidate pools further limit speed comparisons. Small amplitude storage is not total-system memory saving.

Selected runs retain query IDs, model revisions, data hashes, source snapshots, configurations, features, logits, timings, per-seed traces, sessions, and artifact hashes. A manuscript-only exporter verifies inputs, independently recomputes metrics from saved rankings, and generates result tables and outcome-bearing text. NIST checks and tie-corrected statistics remain bound to their original manifests, without rewriting experiments. A central mapping permits replacing a completed full-test source after coverage/provenance verification. Companion records retain all controls, protocol exceptions, commands, and outstanding limitations.

VII. CONCLUSION

We evaluated L6-guided Grover Adaptive Search for selecting compact evidence sets in a hybrid retrieval pipeline. Classical retrieval and a cross-encoder supply candidates and relevance scores, while amplitude amplification searches for improving document subsets in an ideal compact simulator. On the SciFact and NFCorpus follow-up tests, protected-prefix refinement preserved the first five results and reduced mean embedding redundancy relative to matched base selection. The recall changes and their uncertainty are reported explicitly; preserving neural-score thresholds does not prove preservation of judged relevance. The full FiQA evaluation uses the earlier configuration and extends domain and corpus-size coverage, without validating the refinement on that collection. Matched classical optimizers are essential controls: any shared effectiveness benefit belongs to the constrained selection policy rather than uniquely to Grover search. Additional neural scoring and simulation remain part of the complete pipeline cost. The evidence therefore supports a reproducible relevance–redundancy tradeoff, not physical quantum acceleration or monetary savings. Future work should complete the remaining configuration evaluations, implement reversible constraint oracles, and test quantum resources and generated-answer quality independently.

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